

Aditya Dhiman

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Skills

Languages: C, C++, Python, ARM Assembly, SystemVerilog

Embedded: FreeRTOS, STM32, Yocto, BitBake, Linux, MATLAB & Simulink, ROS2

Tools: Vector CANoe, Bazel, JTAG, Logic Analyzer, Altium, CMake, gRPC, Protobuf, GoogleTest, Docker

Education

University of British Columbia
BASC - Computer Engineering

Expected December 2027

Experience

Embedded Software Engineer Intern - AI Platform

Tesla

Palo Alto, CA

June 2026 – Present

- Developing and improving real-time **bare-metal** firmware for Tesla's **Autopilot** compute hardware in **C**, working at the hardware-software interface to optimize system behavior and reliability
- Triaging **MCU reset causes** across production vehicle fleets by analyzing **CAN logs** and real-time **diagnostic logs** from the MCU; identifying root causes and resolving firmware bugs to improve vehicle reliability and Autopilot safety

Software Engineer Intern – Infotainment Platform

Rivian and Volkswagen Group Technologies

Vancouver, BC

Sept 2025 – April 2026

- Developed an **Ethernet** switch manager for the Rivian R2 in **C++**, interfacing with registers for link status, temperature, and heartbeat; debugged command flows over Ethernet using systemd journal logs
- Architected a **multithreaded gRPC server** in C++ streaming Ethernet switch diagnostics to OBD-II clients, implementing Protobuf schemas and bidirectional communication for real-time vehicle diagnostics
- Developed a **GPT partitioning tool** for embedded Linux images integrated into flashing scripts and **CI Pipelines**, replacing a flaky legacy flow and enabling faster, more reliable infotainment ECU flashing for **200+ developers**
- Architected a **crash-resilient** logging framework in C++ with persistent **UFS** storage, automatic log rotation, and cloud upload across ECU power-cycles, reducing post-crash debug time across Infotainment and Connectivity organizations
- Maintaining **Embedded Linux** builds using **Yocto/BitBake**, authoring and debugging recipes for application and BSP layers across current and next-generation infotainment ECUs

Projects

Vehicle Controls Firmware Developer

UBC Formula Electric (FSAE)

Sept 2024 – Present

- Developed and optimized real-time firmware for custom **STM32H5/H7** boards in **C/C++**, configuring **FreeRTOS** tasks, high-speed **CAN/I2C/SPI/UART** buses, and on-chip ADC/GPIO/PWM peripherals; validated hardware bring-up using **JTAG** and logic analyzers to verify bus timing and signal integrity
- Designed a shared, hardware-agnostic **eFuse** driver with current sense-line fault handling and retry logic, using asynchronous **DMA**-driven ADC acquisition to improve reliability of vehicle start-up and fault state management
- Developed a driver for an automotive-grade **IMU** over **SPI** with filtering and FIFO capabilities for acceleration and angular velocity acquisition, necessary for vehicle dynamics control
- Built a **CAN-FD** and Classic CAN driver enabling reliable cross-communication between legacy and next-gen ECUs
- Implemented a **CORDIC** accelerator driver, cutting CPU cycles for trigonometric math by **3×** and reducing torque-vectoring control loop latency
- Implemented a Yaw Rate Controller with **PID** and load-transfer torque allocation for quad-motor torque vectoring; simulated in **Simulink**, deployed in **C**, and validated on vehicle for improved stability and performance
- Developed an **Extended Kalman Filter** in Simulink and C++ fusing IMU, wheel speed, GPS, and slip detection for precise low-speed body velocity estimation below 10 km/h

Roll & Pitch Angle Estimator *C, ESP32, FreeRTOS, Sensor Fusion*

- Developed a robust **IMU** driver using **C** and **FreeRTOS** with sensor fusion algorithms, optimizing data acquisition and processing for low-latency, high-performance communication using I2C between the IMU and ESP32
- Implemented accelerometer **calibration** using offset correction and transformation matrices to compensate for sensor misalignment and bias, improving accuracy of fused angle estimates